1 Use SafeArea Widget in flutter App development

```
import 'package:flutter/material.dart';

void main() {
  runApp(MyApp());
}

class MyApp extends StatelessWidget {
  @override
  Widget build(BuildContext context) {
    return MaterialApp(
      home: SafeArea(
        child: Scaffold(
          appBar: AppBar(
            title: Text("SafeArea Example"),
          ),
          body: Center(
            child: Text("Hello, Flutter!"),
          ),
        ),
      ),
    );
  }
}
```

2 Use AppBar Widget in flutter App development

```
import 'package:flutter/material.dart';

void main() {
  runApp(MyApp());
}

class MyApp extends StatelessWidget {
  @override
  Widget build(BuildContext context) {
    return MaterialApp(
      home: Scaffold(
        appBar: AppBar(
          title: Text("AppBar Demo"),
          leading: Icon(Icons.menu), // Left icon
          actions: [
            IconButton(icon: Icon(Icons.search), onPressed: () {}),
            IconButton(icon: Icon(Icons.more_vert), onPressed: () {}),
          ],
        ),
      ),
```

```
    );
  }
}
```

3 Use Row Widget in flutter App development

```
import 'package:flutter/material.dart';
void main() => runApp(MyApp());

class MyApp extends StatelessWidget {
  @override
  Widget build(BuildContext context) {
    return MaterialApp(
      home: Scaffold(
        appBar: AppBar(title: Text("Row Example")),
        body: Row(
          mainAxisAlignment: MainAxisAlignment.spaceEvenly,
          children: [
            Text("A"), Text("B"), Text("C")
          ],
        ),
      ),
    );
  }
}
```

4 Use Column Widget in flutter App development

```
import 'package:flutter/material.dart';
void main() => runApp(MyApp());

class MyApp extends StatelessWidget {
  @override
  Widget build(BuildContext context) {
    return MaterialApp(
      home: Scaffold(
        appBar: AppBar(title: Text("Column Example")),
        body: Column(
          mainAxisAlignment: MainAxisAlignment.center,
          children: [
            Text("One"), Text("Two"), Text("Three")
          ],
        ),
      ),
    );
  }
}
```

5 Use Container Widget with child property in flutter App development

```
import 'package:flutter/material.dart';
void main() => runApp(MyApp());

class MyApp extends StatelessWidget {
  @override
  Widget build(BuildContext context) {
    return MaterialApp(
      home: Scaffold(
        appBar: AppBar(title: Text("Container Child")),
        body: Center(
          child: Container(
            child: Text("Hello Container"),
          ),
        ),
      ),
    );
  }
}
```

6 Use Container Widget with color property in flutter App development

```
body: Center(
  child: Container(
    color: Colors.blue,
    width: 100,
    height: 100,
  ),
)
```

7 Use Container Widget with height and width property in flutter App development

```
body: Center(
  child: Container(
    height: 150,
    width: 150,
    color: Colors.red,
  ),
)
```

8 Use Container Widget with margin property in flutter App development

```
body: Container(
  margin: EdgeInsets.all(30),
```

```
  color: Colors.green,
)
```

9 Use Container Widget with padding property in flutter App development

```
body: Container(
  padding: EdgeInsets.all(20),
  color: Colors.orange,
  child: Text("Padding Example"),
)
```

10 Use Container Widget with alignment property in flutter App development

```
body: Container(
  alignment: Alignment.center,
  color: Colors.purple,
  child: Text("Centered Text"),
)
```

11 Use Container Widget with decoration property in flutter App development

```
body: Center(
  child: Container(
    height: 100,
    width: 100,
    decoration: BoxDecoration(
      color: Colors.blue,
      borderRadius: BorderRadius.circular(20),
    ),
  ),
)
```

12 Use Container Widget with transform property in flutter App development

```
import 'dart:math';
...
body: Center(
  child: Container(
    transform: Matrix4.rotationZ(pi / 6),
    color: Colors.red,
    padding: EdgeInsets.all(20),
    child: Text("Rotated"),
  ),
)
```

13 Use FlatButton Widget in flutter App development

```
import 'package:flutter/material.dart';

void main() => runApp(MyApp());

class MyApp extends StatelessWidget {
  @override
  Widget build(BuildContext context) {
    return MaterialApp(
      home: Scaffold(
        appBar: AppBar(title: Text("TextButton Example")),
        body: Center(
          child: TextButton(
            onPressed: () {
              print("TextButton Pressed");
            },
            child: Text("Flat Button"),
          ),
        ),
      ),
    );
  }
}
```

14 Use RaisedButton Widget in flutter App development

```
import 'package:flutter/material.dart';

void main() => runApp(MyApp());

class MyApp extends StatelessWidget {
  @override
  Widget build(BuildContext context) {
    return MaterialApp(
      home: Scaffold(
        appBar: AppBar(title: Text("ElevatedButton Example")),
        body: Center(
          child: ElevatedButton(
            onPressed: () {
              print("Elevated Button Pressed");
            },
            child: Text("Raised Button"),
          ),
        ),
      ),
    );
  }
```

```
}
```

15 Use Floating Action Button Widget in flutter App development

```
import 'package:flutter/material.dart';

void main() => runApp(MyApp());

class MyApp extends StatelessWidget {
  @override
  Widget build(BuildContext context) {
    return MaterialApp(
      home: Scaffold(
        appBar: AppBar(title: Text("FAB Example")),
        body: Center(child: Text("Press the button")),
        floatingActionButton: FloatingActionButton(
          onPressed: () {
            print("FAB Clicked");
          },
          child: Icon(Icons.add),
        ),
      ),
    );
  }
}
```

16 Use Dropdown Button Widget in flutter App development

```
import 'package:flutter/material.dart';

void main() => runApp(MyApp());

class MyApp extends StatefulWidget {
  @override
  State<MyApp> createState() => _MyAppState();
}

class _MyAppState extends State<MyApp> {
  String selected = "A";

  @override
  Widget build(BuildContext context) {
    return MaterialApp(
      home: Scaffold(
        appBar: AppBar(title: Text("Dropdown Example")),
        body: Center(
          child: DropdownButton<String>(
            value: selected,
            items: ["A", "B", "C"].map((item) {
```

```
            return DropdownMenuItem(
              value: item,
              child: Text(item),
            );
          }).toList(),
          onChanged: (value) {
            setState(() {
              selected = value!;
            });
          },
        ),
      ),
    ),
  );
 }
}
```

17 Use Icon Button Widget in flutter App development

```
import 'package:flutter/material.dart';

void main() => runApp(MyApp());

class MyApp extends StatelessWidget {
  @override
  Widget build(BuildContext context) {
    return MaterialApp(
      home: Scaffold(
        appBar: AppBar(title: Text("IconButton Example")),
        body: Center(
          child: IconButton(
            icon: Icon(Icons.favorite),
            onPressed: () {
              print("Icon Clicked");
            },
          ),
        ),
      ),
    );
  }
}
```

18 Use RichText Widget in flutter App development

```
import 'package:flutter/material.dart';

void main() => runApp(MyApp());
```

```
class MyApp extends StatelessWidget {
  @override
  Widget build(BuildContext context) {
    return MaterialApp(
      home: Scaffold(
        appBar: AppBar(title: Text("RichText Example")),
        body: Center(
          child: RichText(
            text: TextSpan(
              text: "Hello ",
              style: TextStyle(color: Colors.black, fontSize: 22),
              children: [
                TextSpan(
                  text: "Flutter",
                  style: TextStyle(color: Colors.blue),
                ),
              ],
            ),
          ),
        ),
      ),
    );
  }
}
```

19 Use Container Widget in flutter App development.

```
import 'package:flutter/material.dart';

void main() => runApp(MyApp());

class MyApp extends StatelessWidget {
  @override
  Widget build(BuildContext context) {
    return MaterialApp(
      home: Scaffold(
        appBar: AppBar(title: Text("Container Example")),
        body: Center(
          child: Container(
            height: 100,
            width: 100,
            color: Colors.blue,
            child: Center(child: Text("Box")),
          ),
        ),
      ),
    );
```

```
  }
}
```

20 Use GridView Widget in flutter App development.

```
import 'package:flutter/material.dart';

void main() => runApp(MyApp());

class MyApp extends StatelessWidget {
  @override
  Widget build(BuildContext context) {
    return MaterialApp(
      home: Scaffold(
        appBar: AppBar(title: Text("GridView Example")),
        body: GridView.count(
          crossAxisCount: 2,
          children: List.generate(6, (index) {
            return Container(
              margin: EdgeInsets.all(10),
              color: Colors.blue,
            );
          }),
        ),
      ),
    );
  }
}
```

21 Use ListView Widget in flutter App development.

```
import 'package:flutter/material.dart';

void main() => runApp(MyApp());

class MyApp extends StatelessWidget {
  @override
  Widget build(BuildContext context) {
    return MaterialApp(
      home: Scaffold(
        appBar: AppBar(title: Text("ListView Example")),
        body: ListView(
          children: [
            ListTile(title: Text("Item 1")),
            ListTile(title: Text("Item 2")),
            ListTile(title: Text("Item 3")),
          ],
        ),
      ),
```

```
    );
  }
}
```

22 Use StackView Widget in flutter App development.

```
import 'package:flutter/material.dart';

void main() => runApp(MyApp());

class MyApp extends StatelessWidget {
  @override
  Widget build(BuildContext context) {
    return MaterialApp(
      home: Scaffold(
        appBar: AppBar(title: Text("Stack Example")),
        body: Stack(
          children: [
            Container(color: Colors.blue),
            Positioned(
              top: 100,
              left: 100,
              child: Text("On Top"),
            ),
          ],
        ),
      ),
    );
  }
}
```

23 Use CardView Widget in flutter App development.

```
import 'package:flutter/material.dart';

void main() => runApp(MyApp());

class MyApp extends StatelessWidget {
  @override
  Widget build(BuildContext context) {
    return MaterialApp(
      home: Scaffold(
        appBar: AppBar(title: Text("Card Example")),
        body: Center(
          child: Card(
            elevation: 5,
            child: Padding(
              padding: EdgeInsets.all(20),
              child: Text("Card Widget"),
```

```
          ),
        ),
      ),
    ),
  );
 }
}
```

24 Use ListTile Widget in flutter App development.

```
import 'package:flutter/material.dart';

void main() => runApp(MyApp());

class MyApp extends StatelessWidget {
  @override
  Widget build(BuildContext context) {
    return MaterialApp(
      home: Scaffold(
        appBar: AppBar(title: Text("ListTile Example")),
        body: ListTile(
          leading: Icon(Icons.person),
          title: Text("John Doe"),
          subtitle: Text("Flutter Developer"),
        ),
      ),
    );
  }
}
```

25 Use flutter icons in flutter App development.

```
import 'package:flutter/material.dart';

void main() => runApp(MyApp());

class MyApp extends StatelessWidget {
  @override
  Widget build(BuildContext context) {
    return MaterialApp(
      home: Scaffold(
        appBar: AppBar(title: Text("Icons Example")),
        body: Row(
          mainAxisAlignment: MainAxisAlignment.spaceEvenly,
          children: [
            Icon(Icons.home),
            Icon(Icons.settings),
            Icon(Icons.favorite),
          ],
        ),
```

```
      ),
    );
  }
}
```

26 Use flutter image.asset to add image in flutter App development.
28 Use flutter image.asset to add images in flutter App development.

```
import 'package:flutter/material.dart';

void main() => runApp(MyApp());

class MyApp extends StatelessWidget {
  @override
  Widget build(BuildContext context) {
    return MaterialApp(
      home: Scaffold(
        appBar: AppBar(title: Text("Image Example")),
        body: Center(
          child: Image.asset("assets/image.png"),
        ),
      ),
    );
  }
}
```

27 Use Flutter Navigation and Routing in flutter App development.

```
import 'package:flutter/material.dart';

void main() => runApp(MyApp());

class MyApp extends StatelessWidget {
  @override
  Widget build(BuildContext context) {
    return MaterialApp(home: FirstPage());
  }
}

class FirstPage extends StatelessWidget {
  @override
  Widget build(BuildContext context) {
    return Scaffold(
      appBar: AppBar(title: Text("Page 1")),
      body: Center(
        child: ElevatedButton(
          child: Text("Go to Page 2"),
```

```
        onPressed: () {
          Navigator.push(
            context,
            MaterialPageRoute(builder: (_) => SecondPage()),
          );
        },
      ),
    ),
  );
 }
}

class SecondPage extends StatelessWidget {
 @override
 Widget build(BuildContext context) {
  return Scaffold(
    appBar: AppBar(title: Text("Page 2")),
    body: Center(
      child: ElevatedButton(
        onPressed: () => Navigator.pop(context),
        child: Text("Back"),
      ),
    ),
  );
 }
}
```

29 Use flutter to implement Gesture in App development.

```
import 'package:flutter/material.dart';

void main() => runApp(MyApp());

class MyApp extends StatelessWidget {
 @override
 Widget build(BuildContext context) {
  return MaterialApp(
    home: Scaffold(
      appBar: AppBar(title: Text("Gesture Example")),
      body: Center(
        child: GestureDetector(
          onTap: () {
            print("Tapped!");
          },
          child: Container(
            padding: EdgeInsets.all(20),
            color: Colors.blue,
            child: Text("Tap Me"),
```

```
            ),
          ),
        ),
      ),
    );
  }
}
```

PWA

1 Write manifest.json file and include it in the website.

<u>Index.html</u>

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">

    <!-- Manifest File -->
    <link rel="manifest" href="manifest.json">

    <!-- Theme Color -->
    <meta name="theme-color" content="#4CAF50">

    <title>My PWA Website</title>

    <style>
        body {
            font-family: Arial, sans-serif;
            text-align: center;
            background-color: #f5f5f5;
        }
        h1 {
            color: #4CAF50;
        }
    </style>
</head>
<body>

    <h1>Welcome to My Website</h1>
    <p>This is a Progressive Web App (PWA) example.</p>

    <!-- Service Worker Registration -->
    <script>
        if ('serviceWorker' in navigator) {
            navigator.serviceWorker.register('service-worker.js')
```

```
            .then(function() {
                console.log("Service Worker Registered");
            })
            .catch(function(error) {
                console.log("Service Worker Registration Failed:", error);
            });
        }
    </script>

</body>
</html>
```

2.manifest.json

```
{
  "name": "My Progressive Web App",
  "short_name": "MyPWA",
  "start_url": "index.html",
  "display": "standalone",
  "background_color": "#ffffff",
  "theme_color": "#4CAF50",
  "orientation": "portrait",
  "icons": [
    {
      "src": "icon.png",
      "sizes": "192x192",
      "type": "image/png"
    },
    {
      "src": "icon.png",
      "sizes": "512x512",
      "type": "image/png"
    }
  ]
}
```

3.service-worker.js

```
const CACHE_NAME = "my-site-cache-v1";

const urlsToCache = [
  "index.html",
  "manifest.json",
  "icon.png"
];

self.addEventListener("install", function(event) {
  event.waitUntil(
```

```
    caches.open(CACHE_NAME)
      .then(function(cache) {
        return cache.addAll(urlsToCache);
      })
  );
});

self.addEventListener("fetch", function(event) {
  event.respondWith(
    caches.match(event.request)
      .then(function(response) {
        return response || fetch(event.request);
      })
  );
});
```

2 Write,register, install and activate a service worker file and include it in the website.

`index.html` (Complete)

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">

  <!-- Manifest -->
  <link rel="manifest" href="manifest.json">

  <meta name="theme-color" content="#4CAF50">
  <title>PWA Service Worker Demo</title>

  <style>
      body {
          text-align: center;
          font-family: Arial;
          background: #f5f5f5;
      }
      h1 {
          color: #4CAF50;
      }
  </style>
</head>
```

```
<body>

  <h1>Service Worker Demo</h1>
  <p>This website works offline after first load.</p>

  <!-- Register Service Worker -->
  <script>
    if ('serviceWorker' in navigator) {
      navigator.serviceWorker.register('service-worker.js')
      .then(function(registration) {
        console.log("Service Worker Registered", registration);
      })
      .catch(function(error) {
        console.log("Service Worker Registration Failed", error);
      });
    }
  </script>

</body>
</html>
```

✅ 2. `manifest.json` (Complete)

```
{
 "name": "My PWA Demo",
 "short_name": "PWA Demo",
 "start_url": "index.html",
 "display": "standalone",
 "background_color": "#ffffff",
 "theme_color": "#4CAF50",
 "orientation": "portrait",
 "icons": [
  {
   "src": "icon.png",
   "sizes": "192x192",
   "type": "image/png"
  }
 ]
}
```

👉 ⚠️ Make sure `icon.png` is present OR remove icons section if not available.

✅ 3. `service-worker.js` (Complete)

```
const CACHE_NAME = "my-cache-v1";

const urlsToCache = [
 "index.html",
 "manifest.json"
];

// INSTALL EVENT
self.addEventListener("install", function(event) {
 console.log("Service Worker Installed");

 event.waitUntil(
   caches.open(CACHE_NAME)
     .then(function(cache) {
       return cache.addAll(urlsToCache);
     })
 );
});

// ACTIVATE EVENT
self.addEventListener("activate", function(event) {
 console.log("Service Worker Activated");

 event.waitUntil(
   caches.keys().then(function(cacheNames) {
     return Promise.all(
       cacheNames.map(function(name) {
         if (name !== CACHE_NAME) {
           return caches.delete(name);
         }
       })
     );
   })
 );
});

// FETCH EVENT
self.addEventListener("fetch", function(event) {
 event.respondWith(
   caches.match(event.request)
     .then(function(response) {
       return response || fetch(event.request);
     })
 );
});
```

3 Write a code for static caching in the service worker file and include it in the website.

✅ 1. `index.html`

```html
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Static Caching Demo</title>

  <!-- Manifest -->
  <link rel="manifest" href="manifest.json">

  <!-- CSS -->
  <link rel="stylesheet" href="styles.css">
</head>
<body>

  <h1>Static Caching in Service Worker</h1>
  <p>This website works offline after first load.</p>

  <!-- JS -->
  <script src="script.js"></script>

  <!-- Service Worker Registration -->
  <script>
    if ('serviceWorker' in navigator) {
      navigator.serviceWorker.register('service-worker.js')
      .then(function() {
        console.log("Service Worker Registered");
      })
      .catch(function(error) {
        console.log("SW Registration Failed:", error);
      });
    }
  </script>

</body>
</html>
```

✅ 2. `service-worker.js` (Static Caching)

```
const CACHE_NAME = "static-cache-v1";

// All static files to cache
const STATIC_ASSETS = [
 "/",
 "/index.html",
 "/manifest.json",
 "/styles.css",
 "/script.js"
];

// INSTALL EVENT → Cache all static files
self.addEventListener("install", function(event) {
 console.log("Service Worker Installing...");

 event.waitUntil(
   caches.open(CACHE_NAME)
     .then(function(cache) {
       return cache.addAll(STATIC_ASSETS);
     })
 );
});

// ACTIVATE EVENT → Delete old cache
self.addEventListener("activate", function(event) {
 console.log("Service Worker Activated");

 event.waitUntil(
   caches.keys().then(function(cacheNames) {
     return Promise.all(
       cacheNames.map(function(name) {
         if (name !== CACHE_NAME) {
           return caches.delete(name);
         }
       })
     );
   })
 );
});

// FETCH EVENT → Serve from cache first
self.addEventListener("fetch", function(event) {
 event.respondWith(
```

```
  caches.match(event.request)
    .then(function(response) {
      return response || fetch(event.request);
    })
 );
});
```

✅ 3. `manifest.json`

```
{
 "name": "Static Cache App",
 "short_name": "CacheApp",
 "start_url": "index.html",
 "display": "standalone",
 "background_color": "#ffffff",
 "theme_color": "#4CAF50"
}
```

✅ 4. `styles.css`

```
body {
  font-family: Arial;
  text-align: center;
  background-color: #f2f2f2;
}

h1 {
  color: #4CAF50;
}
```

✅ 5. `script.js`

```
console.log("JavaScript Loaded Successfully");
```

✅ 6. Folder Structure

```
/my-website
  ├── index.html
```

```
├── manifest.json
├── service-worker.js
├── styles.css
├── script.js
```

4.Write a code for dynamic caching in the service worker file and include it in the website.

✅ 1. `service-worker.js` (Dynamic Caching)

```
const CACHE_NAME = "dynamic-cache-v1";

// INSTALL EVENT
self.addEventListener("install", function(event) {
 console.log("Service Worker Installed");
});

// ACTIVATE EVENT
self.addEventListener("activate", function(event) {
 console.log("Service Worker Activated");
});

// FETCH EVENT → Dynamic Caching
self.addEventListener("fetch", function(event) {
 event.respondWith(
   caches.match(event.request).then(function(cachedResponse) {

     // If found in cache → return it
     if (cachedResponse) {
       return cachedResponse;
     }

     // Else fetch from network and cache it
     return fetch(event.request).then(function(networkResponse) {

       return caches.open(CACHE_NAME).then(function(cache) {
         cache.put(event.request, networkResponse.clone());
         return networkResponse;
       });

     }).catch(function() {
       // Optional fallback (offline)
```

```
      return new Response("You are offline");
    });

  })
 );
});
```

✅ 2. `index.html` (Include Service Worker)

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Dynamic Caching Demo</title>
</head>
<body>

<h1>Dynamic Caching Example</h1>
<p>New resources are cached automatically when accessed.</p>

<script>
if ('serviceWorker' in navigator) {
 navigator.serviceWorker.register('service-worker.js')
  .then(function() {
    console.log("Service Worker Registered");
  })
  .catch(function(error) {
    console.log("Registration Failed:", error);
  });
}
</script>

</body>
</html>
```

✅ 3. Folder Structure

```
/my-website
  ├── index.html
  ├── service-worker.js
```

✅ 4. How Dynamic Caching Works (Exam Explanation)

- Files are **not pre-cached**
- When user requests a resource:
    1. Check cache
    2. If not found → fetch from network
    3. Store it in cache
    4. Return response

✅ 5. Output

- First time → loads from network
- Next time → loads from cache
- New files are **automatically cached**

⚠️ Important Points

- No need to list files manually
- Works for dynamic content (API, images, pages)
- Requires localhost or HTTPS

✅ 6. One-Line Conclusion

👉 *Dynamic caching stores resources at runtime when they are requested and reuses them for future requests, improving performance and offline support.*